Networks and Complexity

Solution 19-7

This is an example solution from the forthcoming book Networks and Complexity. Find more exercises at https://github.com/NC-Book/NCB

Ex 19.7: Divided attention [Lvl. 3]

Starting from the idea 'important people have important friends' derive another type of centrality. Proceed as in the case of spectral centrality, but this time nodes divide up their importance, such that for instance a neighbor of a node of degree four, only receives a quarter of the neighbor's importance. Explore where this idea leads you. Consider the example of a 3-chain (o-o-o).

Solution

Let's define v_i again as the importance of node i, then we can write our idea as

$$v_i = \sum_{[ij]} \frac{v_j}{k_j} = \sum_j A_{ij} \frac{v_j}{k_j} \tag{1}$$

where k_j is the degree of node j. So instead of a division by a constant λ we now have a division by k_j. Because this degree depends on the node j we are looking at we can't simply pull it out of the sum like we did with λ. However the equation still seems linear in terms of k so we should be able to express it as a matrix-vector product, which means we should be able to write it in the form

$$v_i = \sum_j B_{ij} v_j \tag{2}$$

where $\mathbf{B}$ is some matrix that we still need to construct. To gain some intuition let's consider the example of a chain of three nodes (o-o-o). The adjacency matrix for such a chain is

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \tag{3}$$

and the degrees are $k_1 = 1$, $k_2 = 2$, $k_3 = 1$. Hence the equation above gives us

$$v_1 = v_2/2 \tag{4}$$
$$v_2 = (v_1 + v_3) \tag{5}$$
$$v_3 = v_2/2 \tag{6}$$

Let's compare this to how these equations would look like with a general $\mathbf{B}$,

$$v_1 = B_{11}v_1 + B_{12}v_2 + B_{13}v_3 \tag{7}$$
$$v_2 = B_{21}v_1 + B_{22}v_2 + B_{23}v_3 \tag{8}$$
$$v_3 = B_{31}v_1 + B_{32}v_2 + B_{33}v_3 \tag{9}$$

By comparing the two equation systems we can now see

$$B_{11} = B_{13} = B_{22} = B_{31} = B_{33} = 0 \tag{10}$$

$$B_{12} = B_{32} = 1/2 \tag{11}$$

$$B_{21} = B_{23} = 1 \tag{12}$$

Hence,

$$\mathbf{B} = \begin{pmatrix} 0 & 1/2 & 0 \\ 1 & 0 & 1 \\ 0 & 1/2 & 0 \end{pmatrix} \tag{13}$$

So this is like the adjacency matrix but every column has been divided by the degree of the corresponding node. As a result the column sum (sum over all elements within a column) is now 1. For the general case we can write

$$\mathbf{B} = \mathbf{A}\mathbf{K}^{-1} \tag{14}$$

where we defined $\mathbf{K}$ as the degree matrix i.e. the matrix which has the node degrees on the diagonal. So for our example network

$$\mathbf{K} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{15}$$

The power -1 on this matrix indicates the matrix inverse and for our diagonal matrix this is simply

$$\mathbf{K}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{16}$$

Multiplying this from the right has the desired effect of normalizing the columns. But let's not get distracted. Whether we use this notation or not, we can now write

$$v_i = \sum_j B_{ij} v_j \tag{17}$$

or equivalently

$$\boldsymbol{v} = \mathbf{B}\boldsymbol{v} \tag{18}$$

where $\mathbf{B}$ is an adjacency matrix where all the columns have been normalized such that they add up to one. The question is now, does this equation actually have a solution? Before we try to answer this generally, it is good to first check this in our example system. We return to

$$v_1 = v_2/2 \tag{19}$$

$$v_2 = v_1 + v_3 \tag{20}$$

$$v_3 = v_2/2 \tag{21}$$

If we substitute $v_1 = v_2/2$ and $v_3 = v_2/2$ in the second equation we get

$$v_2 = \frac{v_2 + v_2}{2} \tag{22}$$

which is obviously true. So $v_1 = v_2/2 = v_3$ is a solution. As always we can scale the vector v as we want, but for example

$$\boldsymbol{v} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \tag{23}$$

is a solution. This tells us that the centrality of each node is directly proportional to its degree. Will it always be like this with this centrality? We know that we can write the ith component of the product $\mathbf{B}\boldsymbol{v}$ as

$$[\mathbf{B}\boldsymbol{v}]_i = \sum_j B_{ij} v_j \tag{24}$$

Now consider what happens when $\boldsymbol{v} = (k_1, k_2, \ldots)^{\mathrm{T}}$ and $B_{ij} = A_{ij} 1/k_j$. Then

$$[\mathbf{B}\boldsymbol{v}]_i = \sum_j B_{ij} v_j = \sum_j A_{ij} \frac{1}{k_j} k_j = \sum_j A_{ij} = k_i \quad \text{for all } i \tag{25}$$

This shows that $\mathbf{A}\mathbf{K}^{-1}\boldsymbol{k} = \boldsymbol{k}$, confirming that the degree vector is always the steady-state solution.

So in summary, the 'divided attention' centrality that we constructed simply recovers the degree centrality.